Materials Characterisations

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Material Characterisation

- **Resilient Modulus**
- It is well known that pavement materials are not elastic. However, under small load compared to the material strength, and if the load is repeated, the deformation of the pavement materials are almost recoverable and proportional to the load and the material can be considered elastic.

$$M_r = \frac{\sigma_d}{\varepsilon_r}$$

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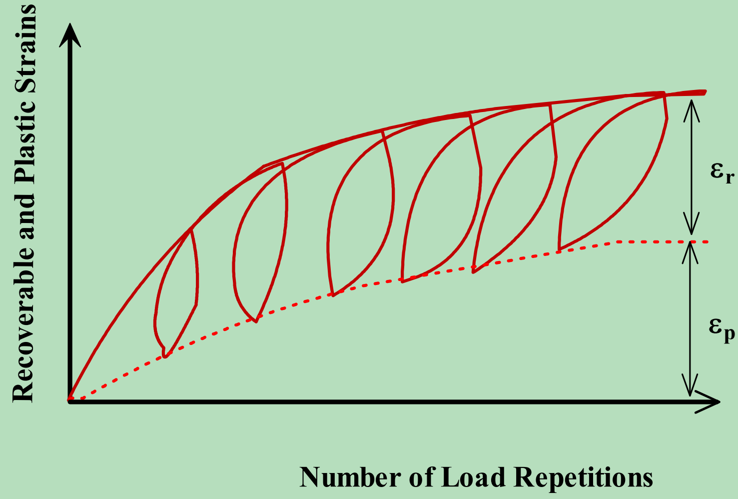

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- σ_d = Deviatoric stress
- ε_r = Recoverable strain
- This test is non-destructive test because the deviator stress is usually small.
- The Loading time can be simply assumed to be inversely proportional to the speed.

$$T = \frac{1}{V}$$

- T (seconds) = Where V is the speed in Km/hr
- Notes: A loading time of 0.1 s and a rest period of 0.9 s can be assumed. The loading time has a very little effect on the resilient modulus of the granular materials, some effect on fine grained soil and a significant effect on bituminous materials.

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Equipment for Measuring the Resilient Modulus

- The resilient modulus of granular materials and fine grained soils can be determined by the repeated load triaxial test.
- **Test Sample** 100 mm (4 in) in diameter and 200 mm (8 in) height
- **Load Cell** is similar to any load cell except it is larger to accommodate the internally mounted load cell and deformation measuring devices.
- The internally mounted load cell and LVDT has advantages over the externally mounted ones where, the friction between the piston and the sample and the equipment deformation can be eliminated in the first.

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- The deformation measuring devices are Linear Variable Differential Transformer (LVDT) attached to the specimen at the upper and lower quarters.
- For Asphalt specimens, the confining pressure has a less effect on the resilient modulus so the unconfined repeated compression test is commonly used. A temperature controlled chamber should be used to maintain the temperature of the specimen.
- The resilient modulus is computed by

$$M_r = \frac{\sigma_d}{\varepsilon_r}$$

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Test Procedure for Granular Materials

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- **Test Procedure**

Sample conditioning

- The sample should subject to the following deviator and confining pressure
- **Confining pressure** **Deviator stress** **N repetition**
- 35 kPa (5 psi) 35 and 69 kPa each for 200
- 69 kPa (10 psi) 69 and 104 kPa each for 200
- 104 kPa 104 and 138 kPa each for 200
- After sample conditioning apply certain confining pressure and a sequence of deviator stress and measure the deformation after 200 load repetitions

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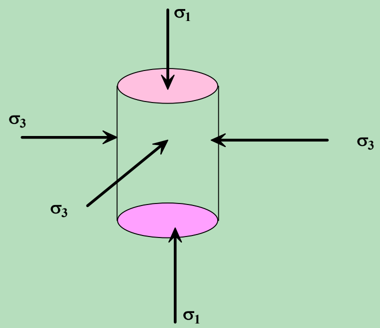

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- Confining Pressure Deviator Stresses

1	138 kPa (20 psi)	(6.9, 14, 35, 104, and 138 kPa) (1,2,5,10, 15, and 20 psi)
2	104 kPa (15 psi)	(6.9, 14, 35, 104, and 138 kPa) (1,2,5,10, 15, and 20 psi)
3	69 kPa (10 psi)	(6.9, 14, 35, and 104 kPa) (1,2,5,10, and 15 psi)
4	35 kPa (5 psi)	(6.9, 14, 35, and 104 kPa) (1,2,5,10, and 15 psi)
5	6.9 kPa (1 psi)	(6.9, 14, 35, 52 and 69 kPa) (1,2,5,7.5, and 10 psi)

- Stop the test after 200 load repetition of the last deviator stress or when the specimen fails.

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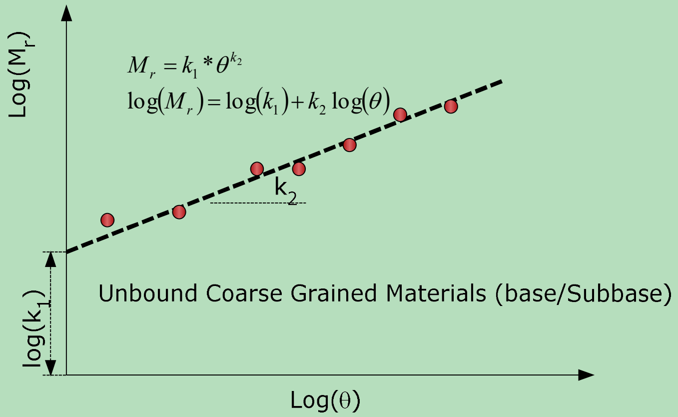
Stress Dependent Behaviour of Unbound Coarse Grained Materials



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Stress Hardening Behaviour

- Coarse grained materials stress hardened with the increase of the confining stresses.
- This behaviour is due to the interlocking between coarse aggregate particles.
- This means the higher the stress the larger the stiffness "higher the modulus" of the materials.

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Resilient Modulus for Fine Grained Soil

- In a similar manner to Coarse grained materials:
- Apply conditioning loading cycles
- Apply a series of confining stresses and for each confining stress, a repeated deviatoric stress is applied and the recoverable deformtion is measured.
- The bi-linear model was found to best model the stress dependent behaviour of the fine grained materials.

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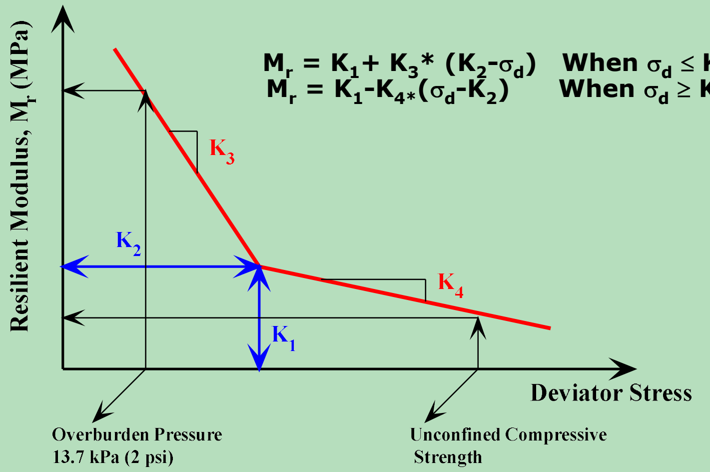
Resilient Modulus - Deviator Stress Model M_r-σ_d Model



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California Bearing Ratio Test (CBR)

- This is an empirical test that is commonly used to characterise the strength of soil (fine or coarse grained, unbound or even lightly stabilised materials)
- It is a penetration test where standard piston of an area 1935 mm^2 (3 in^2), is used to penetrate the soil at standard rate 1.0 mm/min (0.05 in/min) up to 12.7 mm (0.5 in). The ratio between the bearing load of the soil under consideration to the bearing load of standard crushed stone is termed as the California Bearing Ratio CBR.

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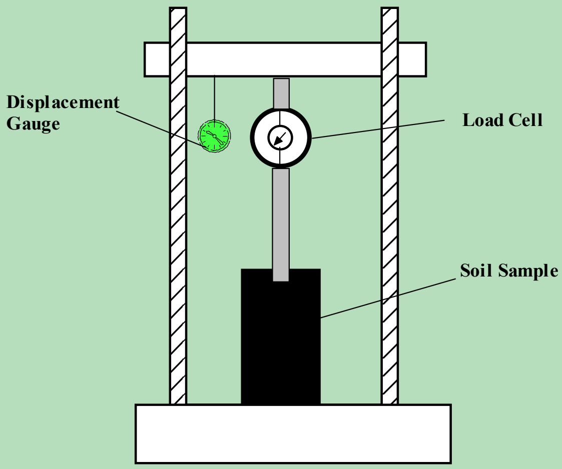
CBR Test



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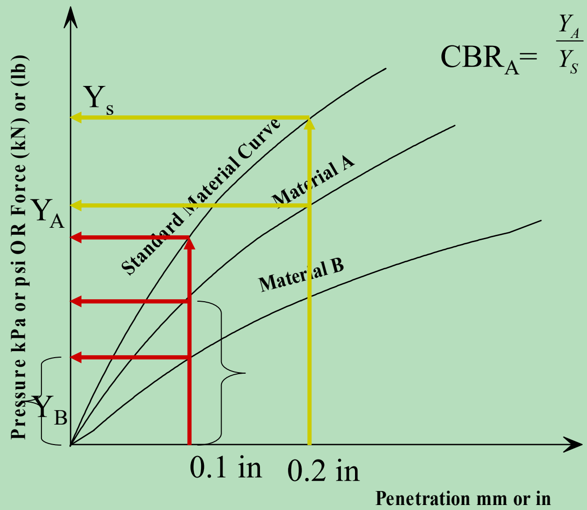
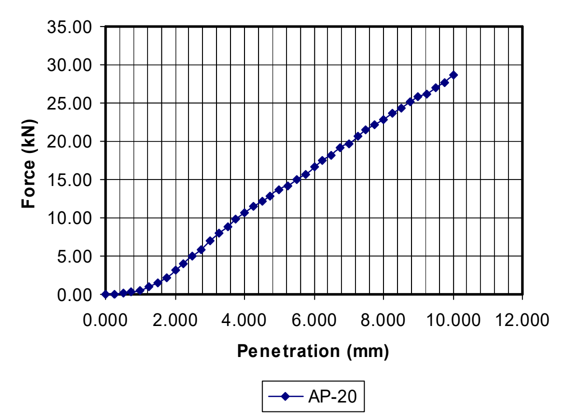
The standard values of high quality crushed stone are as follows:

Penetration	2.5 mm(0.1")	5 mm (0.2")	7.6m(0.3")	10.2mm(0.4")	12.7mm(0.5")
Pressure	6.9 MPa (1000 psi)	10.4 MPa (1500 psi)	13.1 MPa (1900 psi)	15.9 MPa (2300 psi)	17.9 MPa (2600 psi)

Usually the ratio decreases as the penetration increases. The ratio at 2.5 mm (0.1 in) is used as the CBR. In some cases the ratio at 5 mm (0.2 in) is greater than that at 2.5 mm (0.1 in), if this occurs, then the CBR is taken as the ratio at 5 mm (0.2 in).

$$CBR_A = \frac{Y_A}{Y_S} * 100$$





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- **Characterisation of Asphaltic Concrete Mixes**

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Resilient Modulus for Compacted Asphalt Concrete Mix

- The resilient modulus for asphaltic concrete mixture can be done either by uniaxail compression test using specimen 100 mm (4 in) diameter and 200 mm (8 in) height or by using indirect tensile test with specimens 100 mm (4 in) diameter and 64 mm (2.5 in) height. The indirect tensile test is preferred since Marshall specimens or core specimens from the field can be used.

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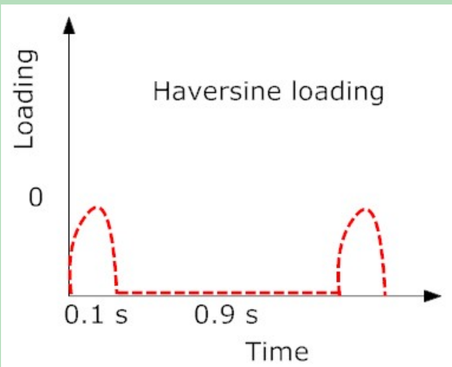

Uniaxial haversine loading with 0.1s loading and 0.9 s rest

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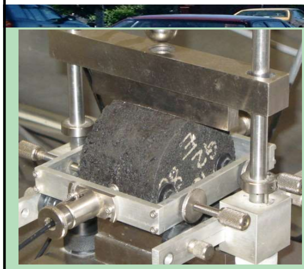
Procedure for Resilient Modulus of Asphalt Mix

- **Sample Conditioning**
- The specimens should be preconditioned until a stable deformation readout is obtained. Usually the number of load repetitions from 50 to 200 is sufficient.

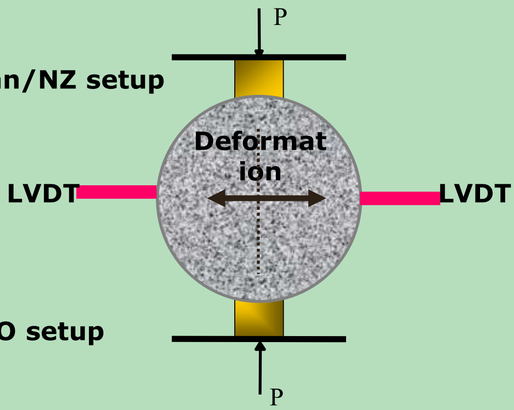
$$E = \frac{\sigma_d}{\varepsilon_r}$$

Indirect Resilient Modulus Test



Australian/NZ setup

AASHTO setup



- In the Australian/NZ setup Poisson ratio is assumed to be 0.4. In the AASHTO setup Poisson ration is measured during the test.
- The resilient modulus should be determined at a range of frequencies and temperatures which are expected to be encountered in the field.
- The data of resilient modulus at different temperatures and frequencies can be used to construct the master curve.

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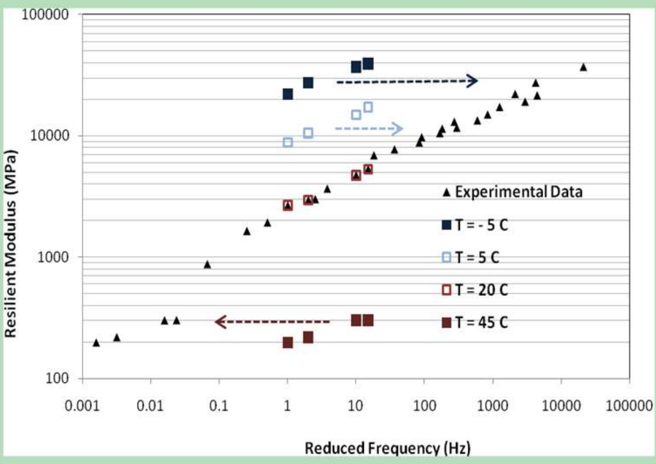
Master Curve using Indirect Resilient modulus data



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$$M_r = \frac{P*(\nu + 0.2734)}{\Delta * h}$$

(This equation for indirect tensile test only)
P = The magnitude of the dynamic load
ν = Poisson ratio
Δ = Total recoverable deformation
h = Specimen thickness

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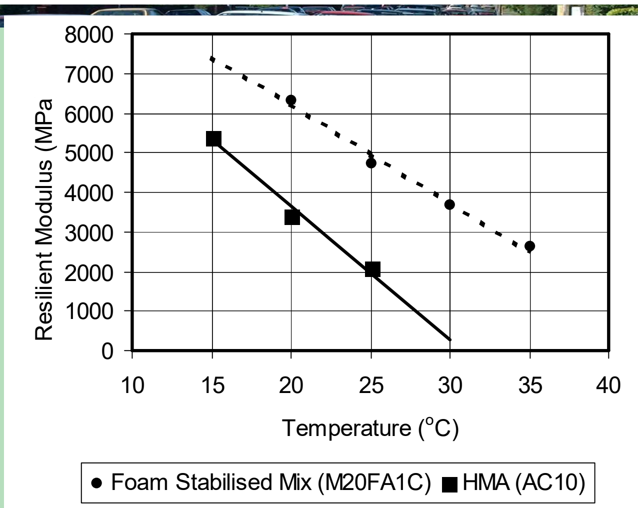

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Dynamic Modulus Test

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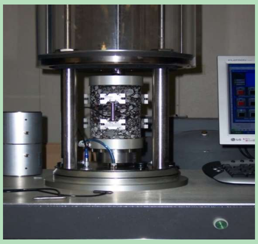

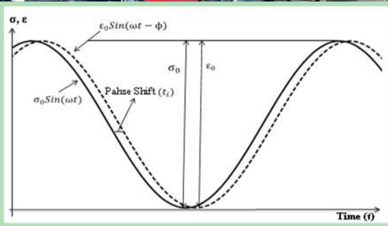

$$|E^*| = \frac{\sigma_0}{\varepsilon_0}$$

$$E^* = E' + iE''$$

$$\Phi = \frac{t_i}{t_p} * 360$$

E’= True part of the complex modulus= Storage or Elastic modulus component

E”= Imaginary part of the complex modulus= Loss or Viscous modulus component

ϕ= Phase angle

Master Curve-Sigmoidal Function

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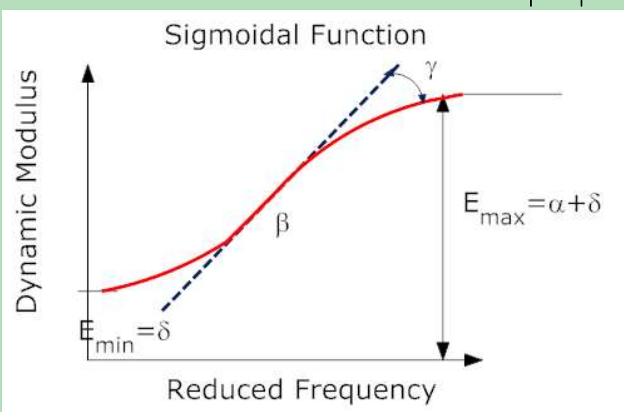
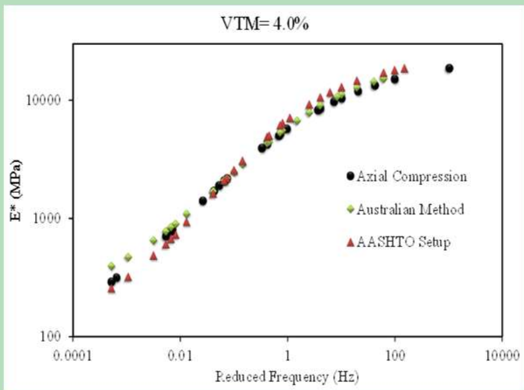
$$\log |E^*| = \delta + \frac{\alpha}{1 + e^{\beta + \gamma(\log f_r)}}$$



Dynamic Modulus Master Curve

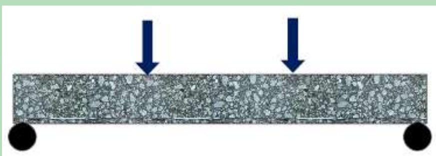


Sigmoidal Function

$$\log \left| E^{*} \right| = \delta + \frac{\alpha}{1 + e^{\beta + \gamma (\log f_r)}}$$

δ= min. modulus
δ+α= max. Modulus

Flexural Modulus for Bituminous Mixtures



- Repeated Sinsoidal loading is applied and the flexural modulus is applied after the 50^{th} repetition. Usually is used for fatigue modelling

Permanent Deformation Characterisation

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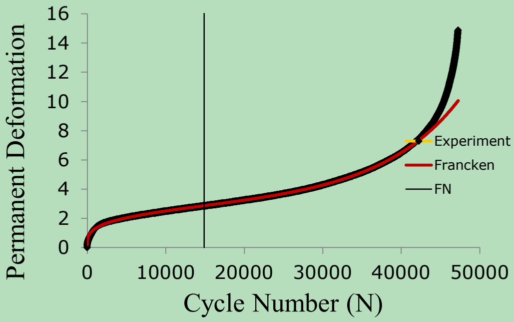
Dynamic Creep Test



– **Flow Number at point of inflection**

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Static Creep Test

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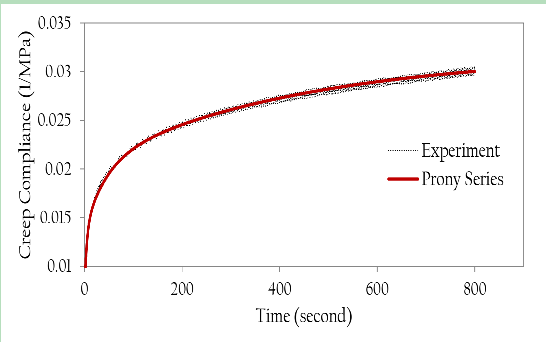
- Apply constant stress and measure permanent deformation with time.



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Wheel Tracker

Wheel load 700 N
Speed is 26.5 Cycle per minutes
Constant Temperature 50/60ºC

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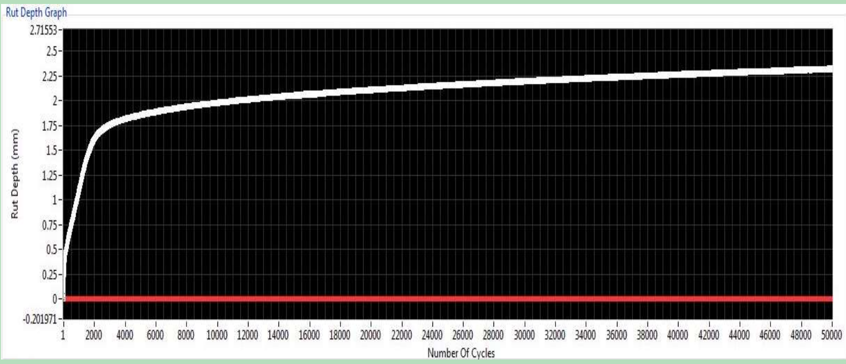
Conventional Wheel tracker test results



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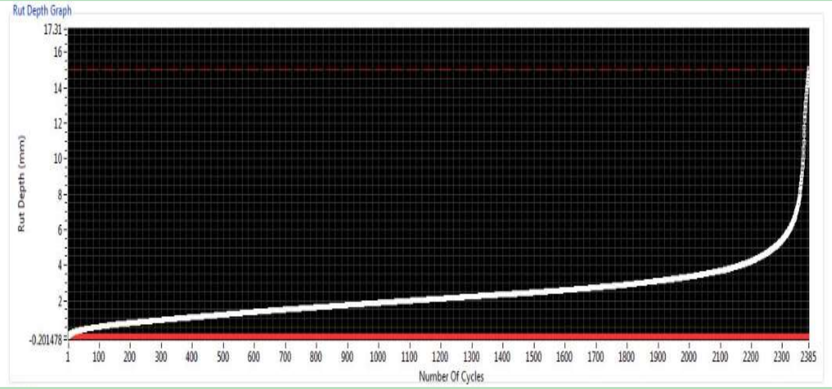
Modified wheel Tracker



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Permanent Deformation Modelling

- Power Model

$$\varepsilon_p = A * N^B$$

- Francken Model

$$\varepsilon_p = A * N^B + C * \left(e^{DN} - 1\right)$$

- Permanent to resilient strain ratio

$$Log\left(\frac{\varepsilon_p}{\varepsilon_r}\right) = A + B * Log(T) + C * Log(N)$$

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Dynamic Modulus of Asphalt and Asphalt Mixtures by Shell Nomographs

- In the absence of actual test data, Shell nomographs can be used to predict the dynamic modulus of the asphalt mix if the aggregate and bitumen volume concentration and the properties of bitumen are known.
- Two nomographs are used, the first nomograph is to determine the stiffness modulus of bitumen and the second nomograph for the determination of the stiffness modulus for the mix.
- The properties of bitumen required are:
- Penetration index (PI): The penetration index expresses the temperature susceptibility of the bitumen. The higher the PI the lower the temperature susceptibility.

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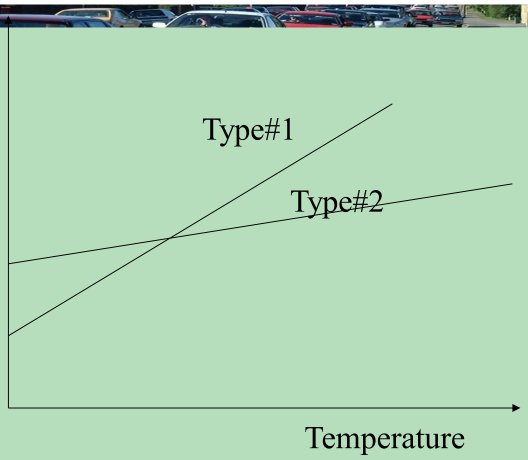
Log (Pen)



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$$A = \frac{Log(Pen_2) - Log(Pen_1)}{T_2 - T_1} = \frac{Log\left(\frac{Pen_2}{Pen_1}\right)}{(T_2 - T_1)}$$

$$PI = \frac{20 - 500A}{1 + 50A}$$

The softening point (Ring and Ball) temperature is convenient reference temperature to determine. The penetration of all asphalts at $T_{R\&B}$ is equal 800.

The temperature to be used in the chart is the normalized temperature which is the difference between the testing temperature and the softening point (Ring and Ball) temperature

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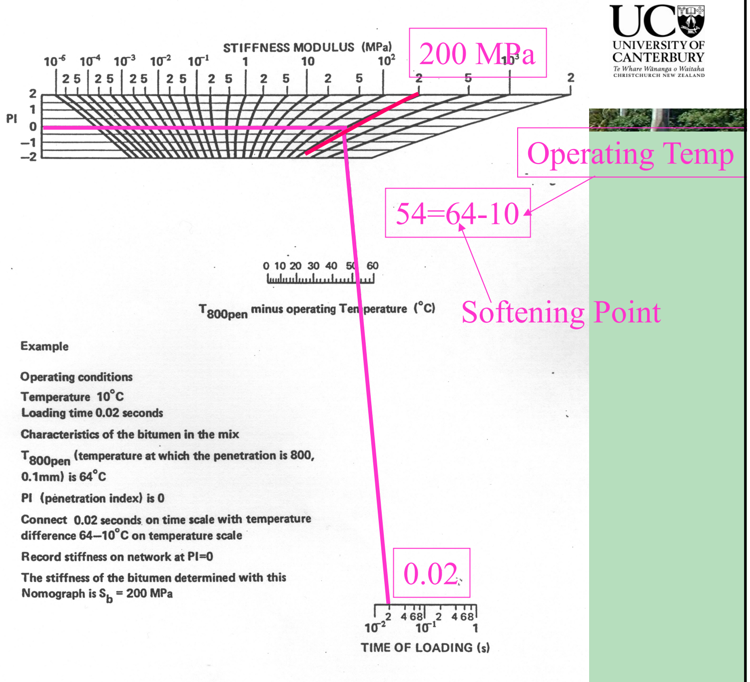

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Factors Affecting M_r

- Bitumen Type
- Bitumen Content
- Air Voids
- Aggregate Type and Gradation
- Temperature
- Rate of Loading

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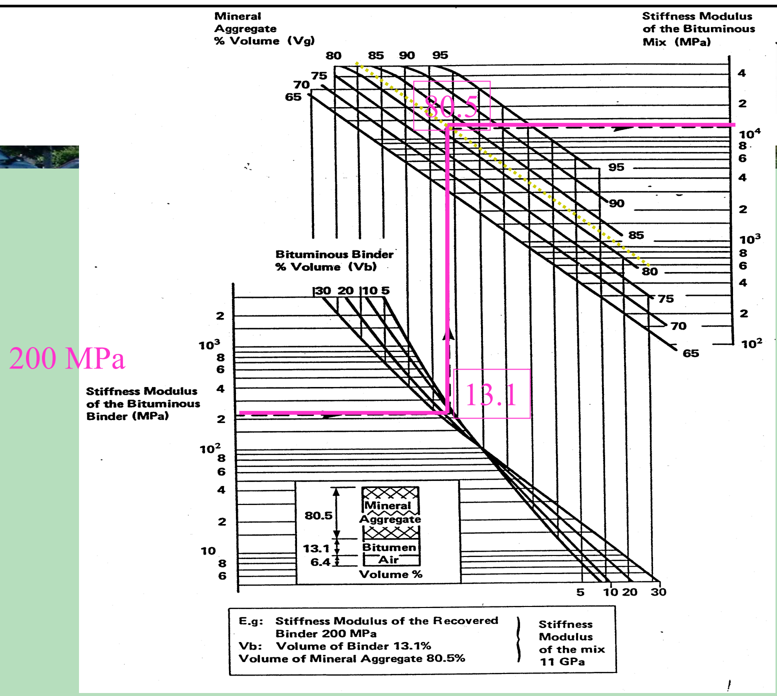

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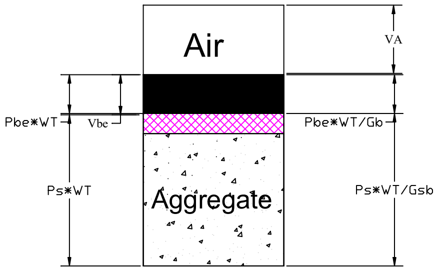

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Factors Affecting Fatigue Life

- Method Used in Measuring the Fatigue Life
- Stiffness of the Mix
- Traffic Loading Spectrum
- Temperature Variation During Service Life

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Laboratory Methods for Fatigue Measurements

- Constant Stress Method
- Constant Stain Method

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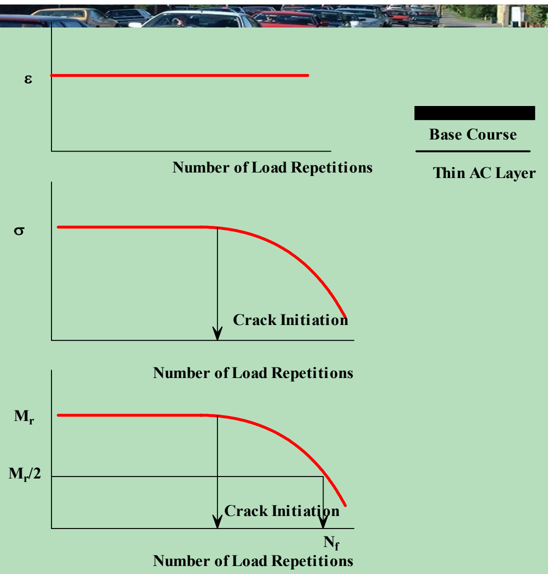
Constant Strain Test



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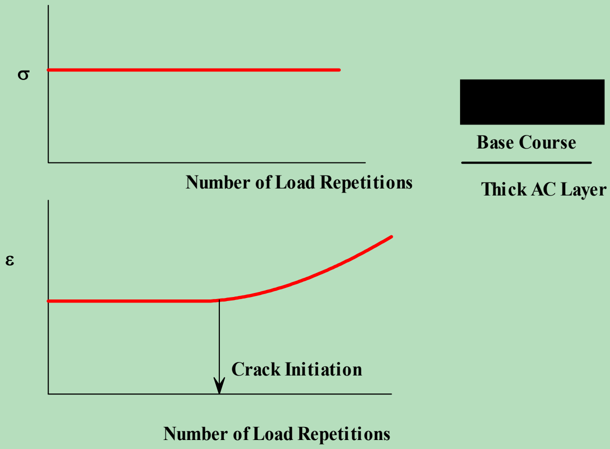
Constant Stress Method



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Laboratory Fatigue Test

- Flexural Test
- Indirect Tensile Test

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- Suggested Fatigue Criteria

$$N_f = \left[\frac{6918 * (0.856 * V_b + 1.08)}{S_{mix}^{\ 0.36} * \varepsilon} \right]^5$$

N_f = Number of Load to Failure
ε = Tensile Strain in µm
V_b = Bitumen Percentage by Volume
S_{mix} = Mix Stiffness (Modulus) MPa

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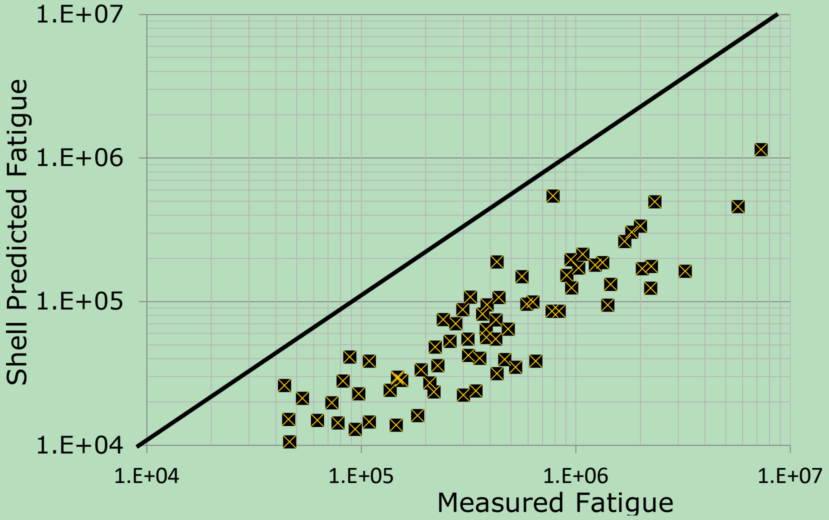
Shell is severely Underestimating fatigue life



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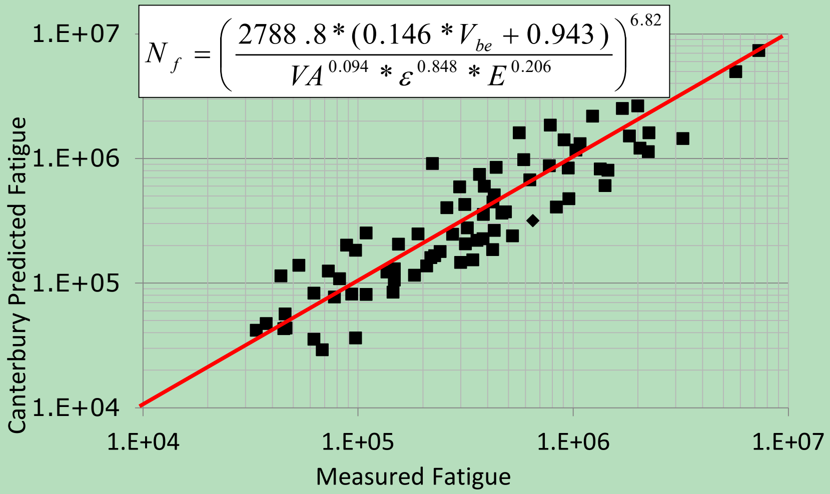
Canterbury Proposed New Fatigue Model

$$N_f = \left(\frac{2788.8 * (0.146 * V_{be} + 0.943)}{VA^{0.094} * \varepsilon^{0.848} * E^{0.206}} \right)^{6.82}$$



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Shortcomings of the Shell Model

- The percent of binder by volume is for the total binder content not the effective binder content. This make no account for bitumen absorption.
- The Shell model is laboratory developed model and it has never been field calibrated according to evidence from research results.

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Evidence from the Shell Manual

4. Correlation between laboratory fatigue tests and practice

The fatigue data obtained in the laboratory cannot be applied directly to thickness design since in practice the mode of loading (random pulse loading with rest periods) and spectrum of strain values (random distribution due to different loads and lateral distribution of loads) are diffe-rent from laboratory conditions (usually sinusoidal load-ing and fixed strain or stress level during one test). There are also indications that some healing occurs in practice and that intermittent loading has a less damaging effect than continuous loading. For these effects correction fac-tors for the design life should be applied. Allowance should also be made for the transverse distribution of wheels loads. In addition, the fatigue life of a pavement influenced by the temperature variations in the asphalt layer. Since both the stiffness modulus of the asphalt mix

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- The Shell model was developed based on 30 mm x 30 mm beams which is quite thin cross section for mixes such as AC14 and AC20 and this is expected to produce very conservative results compared to thicker beams (60 mm x 60 mm)
- The feeling between many practitioners is that the Shell fatigue model does not fit well with our current practice and needs thorough research and reconsideration.

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Mix Types and Properties Used In the Shell Model

Mix no.		1	2	3	4	5	6	7
Mix type		Dense asphaltic concrete	Dense bitumen macadam	Gravel sand asphalt	Lean bitumen macadam	Rolled asphalt, base course mix	French "Grave bitume"	Bitumen sand, base course
Bitumen Grade		40/50	80/100	45/60	80/100	40/60	40/50	45/60
Mix properties								
V_g,	%v	84.1	85.6	78.0	61.9	83.7	81.4	70.8
V_b,	%v	14.2	11.0	11.0	4.9	14.1	9.3	8.9
VIM,	%v	1.7	3.4	11.0	33.2	2.2	9.3	20.3
VMA,	%v	15.9	14.4	22.0	38.1	16.3	18.6	29.2
VFB,	%	89.3	76.4	50.0	13.0	86.5	50.0	30.5

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